INCH-POUND

MIL-C-11272/17B
7 February 2003
SUPERSEDING
MIL-C-11272/17A(USAF)
6 July 1973

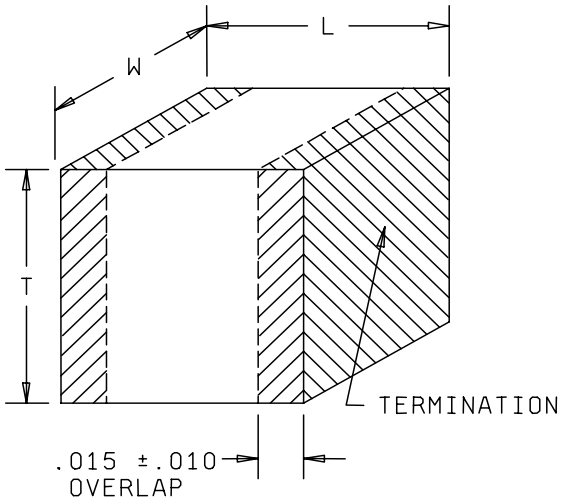
MILITARY SPECIFICATION

CAPACITORS, FIXED, PORCELAIN DIELECTRIC,

STYLES CY83 AND CY84

This specification is approved for use by all Departments and Agencies of the Department of Defense.

INACTIVE FOR DESIGN AFTER
16 JULY 1979. USE MIL-PRF-55681/4.



Inches	mm
.010	.25
.015	.38
.020	.51
.060	1.52
.100	2.54
.110	2.79

Dimensions				
			T	
Style	L	W	Min.	Max.
CY83	.110 ±.015	.110 ±.015	.060	.100
CY84	.110 ±.020	.110 ±.015	.060	.100

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Capacitors, fixed, porcelain dielectric, styles CY83 and CY84.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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TABLE I. Styles CY83 and CY84.

PIN 1/	Capacitance	Rated voltage	Capacitance tolerance available
	pF	Volts, dc	
CY8-D0R1B	0.1	500	B
CY8-D0R2B	0.2	500	B
CY8-D0R3-	0.3	500	B, C
CY8-D0R4-	0.4	500	B, C
CY8-D0R5-	0.5	500	B, C, D
CY8-D0R6-	0.6	500	B, C, D
CY8-D0R7-	0.7	500	B, C, D
CY8-D0R8-	0.8	500	B, C, D
CY8-D0R9-	0.9	500	B, C, D
CY8-D1R0-	1.0	500	B, C, D
CY8-D1R1-	1.1	500	B, C, D
CY8-D1R2-	1.2	500	B, C, D
CY8-D1R3-	1.3	500	B, C, D
CY8-D1R4-	1.4	500	B, C, D
CY8-D1R5-	1.5	500	B, C, D
CY8-D1R6-	1.6	500	B, C, D
CY8-D1R7-	1.7	500	B, C, D
CY8-D1R8-	1.8	500	B, C, D
CY8-D1R9-	1.9	500	B, C, D
CY8-D2R0-	2.0	500	B, C, D
CY8-D2R1-	2.1	500	B, C, D
CY8-D2R2-	2.2	500	B, C, D
CY8-D2R4-	2.4	500	B, C, D
CY8-D2R7-	2.7	500	B, C, D
CY8-D3R0-	3.0	500	B, C, D
CY8-D3R3-	3.3	500	B, C, D
CY8-D3R6-	3.6	500	B, C, D
CY8-D3R9-	3.9	500	B, C, D
CY8-D4R3-	4.3	500	B, C, D
CY8-D4R7-	4.7	500	B, C, D
CY8-D5R1-	5.1	500	B, C, D
CY8-D5R6-	5.6	500	B, C, D
CY8-D6R2-	6.2	500	B, C, D
CY8-D6R8-	6.8	500	B, C, J, K, M
CY8-D7R5-	7.5	500	B, C, J, K, M
CY8-D8R2-	8.2	500	B, C, J, K, M
CY8-D9R1-	9.1	500	B, C, J, K, M
CY8-D100-	10	500	F, G, J, K, M
CY8-D110-	11	500	F, G, J, K, M
CY8-D120-	12	500	F, G, J, K, M
CY8-D130-	13	500	F, G, J, K, M
CY8-D150-	15	500	F, G, J, K, M
CY8-D160-	16	500	F, G, J, K, M
CY8-D180-	18	500	F, G, J, K, M
CY8-D200-	20	500	F, G, J, K, M
CY8-D220-	22	500	F, G, J, K, M
CY8-D240-	24	500	F, G, J, K, M
CY8-D270-	27	500	F, G, J, K, M

See footnote at end of table.

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FIGURE 1. Styles CY83 and CY84 - Continued.

PIN 1/	Capacitance	Rated voltage	Capacitance tolerance available
CY8-D300-	30	500	F, G, J, K, M
CY8-D330-	33	500	F, G, J, K, M
CY8-D360-	36	500	F, G, J, K, M
CY8-D390-	39	500	F, G, J, K, M
CY8-D430-	43	500	F, G, J, K, M
CY8-D470-	47	500	F, G, J, K, M
CY8-D510-	51	500	F, G, J, K, M
CY8-D560-	56	500	F, G, J, K, M
CY8-D620-	62	500	F, G, J, K, M
CY8-D680-	68	500	F, G, J, K, M
CY8-D750-	75	500	F, G, J, K, M
CY8-D820-	82	500	F, G, J, K, M
CY8-D910-	91	500	F, G, J, K, M
CY8-D101-	100	500	F, G, J, K, M
CY8-D111-	110	300	F, G, J, K, M
CY8-D121-	120	300	F, G, J, K, M
CY8-D131-	130	300	F, G, J, K, M
CY8-D151-	150	300	F, G, J, K, M
CY8-D161-	160	300	F, G, J, K, M
CY8-D181-	180	300	F, G, J, K, M
CY8-D201-	200	300	F, G, J, K, M
CY8-D221-	220	200	F, G, J, K, M
CY8-D241-	240	200	F, G, J, K, M
CY8-D271-	270	200	F, G, J, K, M
CY8-D301-	300	200	F, G, J, K, M
CY8-D331-	330	200	F, G, J, K, M
CY8-D361-	360	200	F, G, J, K, M
CY8-D391-	390	200	F, G, J, K, M
CY8-D431-	430	200	F, G, J, K, M
CY8-D471-	470	200	F, G, J, K, M
CY8-D511-	510	100	F, G, J, K, M
CY8-D561-	560	100	F, G, J, K, M
CY8-D621-	620	100	F, G, J, K, M
CY8-D681-	680	50	F, G, J, K, M
CY8-D751-	750	50	F, G, J, K, M
CY8-D821-	820	50	F, G, J, K, M
CY8-D911-	910	50	F, G, J, K, M
CY8-D102-	1000	50	F, G, J, K, M

1/ Complete PIN will include an additional digit to indicate the style, and an additional letter symbol to indicate the capacitance tolerance, where applicable.

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REQUIREMENTS

Requirements: Requirements shall be in accordance with MIL-C-11272, and as specified herein.

Design and construction: Capacitors shall be of the design, construction, and physical dimensions specified on figure 1 and in table I.

Case type: Multi-layer, unencapsulated, monolithic.

Material - Porcelain.

Terminations:

Style CY83: Metallized.

Style CY84: Solder coated, 372°F, 62 Sn, 36P6, 2 percent Ag.

Inspection conditions: All visual examinations shall be performed using a stereo microscope having a minimum magnification of 20 power.

Capacitance (Cap.) value: See table I.

Capacitance tolerance: B: 0.10pF, C: 0.25pF, D: 0.50pF, F: 1%, G: 2%, J: 5%, K: 10%, M: 20%.

Rated voltage: See table I, characteristic D.

Operating temperature range: -55°C to +125°C.

Insulation resistance (IR):

At +25°C: 10^6 megohms, minimum (for 0.1 pF through 470 pF).
10^5 megohms, minimum (for 510 pF through 1,000 pF).

At +125°C: 10^5 megohms, minimum (for 0.1 pF through 470 pF).
10^4 megohms, minimum (for 510 pF through 1,000 pF).

Quality factor: 10,000. 1 MHz -0 +100 kHz.

Life:

At +25°C and +125°C:

IR: Not less than initial requirement.

Cap.: Change not more than 0.2 percent or 0.2 pF, whichever is greater, from the nominal value.

Temperature coefficient and capacitance drift: See table II.

TABLE II. Temperature coefficient and capacitance drift.

Temperature coefficient	Capacitance drift (-55°C to +125°C)
Parts/million/°C 90 ±20	0.1 percent or 0.1 pF, whichever is greater

NOTE: These capacitors are intended for use in RF solid-state circuitry up to and including 25 gigahertz, and cannot be replaced in the field.

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Air Force - 99
DLA - IS

Preparing activity:
DLA - CC

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